AirSentinel PRD

AirSentinel — ESP32 BLE Environmental Safety Node

Product Requirements Document v1.0

Hackathon: Embedded SF: CraftifAI Physical AI Hackathon (HackwithBangalore 4.0) — Track B (FirmGen)
Target: ESP32 dev board (ESP-IDF v5.5), BLE + GPIO + UART **Bonus targeted:** Live Intelligence (telemetry / RTOS monitor)

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1. Problem

Heat, humidity and sensor failures in server closets, labs, greenhouses and storage rooms go unnoticed until equipment or goods are damaged. People need a cheap always-on node that measures the environment, shows its status locally in a way that cannot be misread, alerts audibly, and streams readings to a phone or laptop over BLE with no infrastructure.

2. Product summary

AirSentinel samples temperature and humidity once per second, classifies the room as **OK / WARN / ALARM / FAULT** with debounce and hysteresis (so it never flaps), drives a traffic-light LED set and a buzzer, lets the user acknowledge an alarm with a button, and publishes readings and state over a BLE GATT service. All behaviour is logged over UART in a fixed format.

3. Hardware (confirm pins before freezing)

Item	Default
Board	ESP32 DevKit (ESP32-WROOM-32)
Temp/humidity sensor	DHT22 (AM2302), data on GPIO4
Green / Yellow / Red LEDs	GPIO25 / GPIO26 / GPIO27 (active high, with resistors)
Buzzer (active)	GPIO14 (active high)
Acknowledge button	GPIO32, to GND, internal pull-up
Status LED	On-board LED GPIO2 (BLE state)
UART console	UART0, 115200 8N1

If no DHT22 is available, the sensor driver may be swapped for BME280 on I2C (SDA GPIO21, SCL GPIO22) without changing any other requirement.

4. Functional requirements

4.1 Sensing

- **REQ-SENS-01** The firmware SHALL sample temperature and relative humidity once every 1000 ms (±50 ms).
- **REQ-SENS-02** Temperature SHALL be reported in 0.01 °C resolution and humidity in 0.01 %RH resolution.

- **REQ-SENS-03** A sample SHALL be marked invalid if the sensor read fails or returns a value outside –40… 80 °C or 0…100 %RH.
- **REQ-SENS-04** After 3 consecutive invalid samples the system SHALL enter the FAULT state; one valid sample SHALL leave FAULT (re-evaluating normally).

4.2 State machine

States: `OK`, `WARN`, `ALARM`, `FAULT`. - **REQ-STATE-01** WARN is entered when temperature ≥ 30.0 °C OR humidity ≥ 70 %RH. - **REQ-STATE-02** ALARM is entered when temperature ≥ 35.0 °C. - **REQ-STATE-03** *Debounce:* a transition to a more severe state SHALL occur only after 3 consecutive valid samples support it (≈3 s). - **REQ-STATE-04** *Hysteresis:* a transition to a less severe state SHALL require the value to fall 1.0 °C (or 3 %RH for humidity) below the threshold that caused the current state, for 3 consecutive valid samples. - **REQ-STATE-05** Every state change SHALL be logged once on UART (see REQ-LOG-01) and reflected on outputs within 200 ms of the transition.

4.3 Local indication

- **REQ-IND-01** OK → green LED solid. WARN → yellow LED solid. ALARM → red LED solid. FAULT → all three LEDs blinking together at 2 Hz (250 ms on / 250 ms off). Exactly one indication pattern is active at a time.
- **REQ-IND-02** In ALARM the buzzer SHALL pulse at 2 Hz (250 ms on / 250 ms off) until acknowledged or the state leaves ALARM.
- **REQ-IND-03** Pressing the button (debounced 50 ms, falling edge) while in ALARM SHALL silence the buzzer for 60 s. The red LED stays on. If still in ALARM after 60 s the buzzer resumes.
- **REQ-IND-04** Leaving ALARM SHALL stop the buzzer immediately and clear any acknowledgement.

4.4 BLE

- **REQ-BLE-01** The device SHALL advertise as `AirSentinel-XXXX` where XXXX is the last 2 bytes of the BT MAC in hex, starting within 2 s of boot, and restart advertising after disconnect.
- **REQ-BLE-02** The device SHALL expose a custom GATT service, UUID `7d1a0001-5a3c-4f6e-9b1a-0c5e5e5e0001`.
- **REQ-BLE-03** Characteristic **Reading** (UUID `7d1a0002-5a3c-4f6e-9b1a-0c5e5e5e0001`, Read + Notify) SHALL hold an 8-byte little-endian packet:

Offset	Type	Field
0	int16	temperature, 0.01 °C (0x7FFF if invalid)
2	uint16	humidity, 0.01 %RH (0xFFFF if invalid)
4	uint8	state: 0=OK, 1=WARN, 2=ALARM, 3=FAULT
5	uint8	flags: bit0 = buzzer silenced by ack, bit1 = sample invalid
6	uint16	sequence counter, increments per sample, wraps at 65535

- **REQ-BLE-04** When a client has enabled notifications, the device SHALL notify once per sample (1 Hz).
- **REQ-BLE-05** Characteristic **Ack** (UUID `7d1a0003-5a3c-4f6e-9b1a-0c5e5e5e0001`, Write) SHALL accept a single byte `0x01` and behave exactly as a button press (REQ-IND-03).
- **REQ-BLE-06** The on-board status LED SHALL blink at 1 Hz while advertising and be solid while connected.

4.5 Logging and telemetry (Live Intelligence)

- **REQ-LOG-01** On every sample the firmware SHALL print one UART line: `[SAMPLE] seq=<n> t=<°C, 1 dp> h=<%RH, 1 dp> state=<OK|WARN|ALARM|FAULT> valid=<0|1>`.
- **REQ-LOG-02** On every state change: `[STATE] <old>-><new> t=<°C> h=<%RH>`. On ack: `[ACK] source=<BUTTON|BLE>`.
- **REQ-LOG-03** Runtime telemetry (FirmGen Enable Telemetry) SHALL expose per-task stack high-water marks, free heap, and CPU load so they can be shown live during the demo.

4.6 Reliability

- **REQ-REL-01** The task watchdog SHALL be enabled with a 5 s timeout covering the sampling task; a hang SHALL reset the device.
- **REQ-REL-02** Free heap SHALL not decrease by more than 2 KB over 30 minutes of operation with a connected notifying client.
- **REQ-REL-03** Boot to first indication (LED) SHALL take less than 3 s.

5. Non-functional

- **NFR-01** Language C, ESP-IDF v5.5, FreeRTOS tasks (sampling, indication, BLE) with clear separation; the state machine logic SHALL be in a hardware-independent module so it can be unit tested.
- **NFR-02** No dynamic allocation after initialization in the sampling path.
- **NFR-03** Thresholds and timings SHALL be `#defined`/Kconfig constants in one header.

6. Acceptance criteria (verified on the board)

ID	Criterion	Evidence	Covers
AC-01	Boot, serial shows `[SAMPLE]` line once per second, LED green within 3 s	UART capture + photo	SENS-01, REL-03, IND-01
AC-02	Warm the sensor (hand/hair-dryer) to ≥ 30 °C: yellow after 3 samples, `[STATE] OK->WARN` logged once	UART + photo	STATE-01/03/05
AC-03	Continue to ≥ 35 °C: red + buzzer pulsing, `[STATE] WARN->ALARM`	UART + video	STATE-02, IND-01/02
AC-04	Press button: buzzer silent, red stays on, `[ACK] source=BUTTON`; buzzer resumes after 60 s	UART + video	IND-03
AC-05	Cool below 34.0 °C for 3 samples: returns to yellow, buzzer off. Hovering at 34.5 °C does NOT flap	UART	STATE-04, IND-04
AC-06	Unplug sensor data wire: after 3 s all LEDs blink 2 Hz, `FAULT`; replug: recovers	UART + video	SENS-04, IND-01
AC-07	Phone/laptop sees `AirSentinel-XXXX`, connects, notifications arrive at 1 Hz with the packet decoded correctly	dashboard screenshot	BLE-01…04
AC-08	Writing 0x01 to Ack characteristic from dashboard silences buzzer	dashboard + UART	BLE-05

ID	Criterion	Evidence	Covers
AC-09	Telemetry panel shows stack watermarks, free heap and CPU load live	screenshot	LOG-03
AC-10	Hang injected in sampling task resets the board within 5 s	UART reset banner	REL-01

7. Out of scope

Wi-Fi, OTA, persistent storage, BLE bonding/encryption (documented as future work).